

1. Critical Values from the t-table (Table D).

- (a) **Accurate to the nearest 3 decimal places**, what is the critical value ($t_{\frac{\alpha}{2}, n-1}$ or $z_{\frac{\alpha}{2}}$) that corresponds to the given confidence levels and degrees of freedom. Fill in the following table with the appropriate critical values:

Confidence level	σ unknown ($t_{\frac{\alpha}{2}, n-1}$)			σ known ($z_{\frac{\alpha}{2}}$)
	$df = 6$	$df = 30$	$df = 100$	
80%				
90%				
95%				
99.9%				

- (b) **[Free Response]** As the degrees of freedom increase, how does the $t_{\frac{\alpha}{2}, n-1}$ value compare to the $z_{\frac{\alpha}{2}}$ value?
- (c) A 90% confidence interval using $df = 6$ is based on how many observations?
- (d) A 95% confidence interval using $df = 30$ is based on how many observations?
- (e) **[Don't forget how to use Table A]** What is the positive value of $z_{\frac{\alpha}{2}}$ for a 74% confidence interval when σ is known?

2. Probabilities from the t-table (Table D). Notice that some of the questions below use df and some use n . What is the value of

- (a) $P(T < 3.326)$ for $df = 14$? [up to the nearest 4 decimal places]
- (b) $P(T < -0.859)$ for $n = 22$? [up to the nearest 1 decimal place]
- (c) $P(T > 3.421)$ for $df = 27$? [up to the nearest 3 decimal places]
- (d) $P(T > 3.421)$ for $n = 28$? [up to the nearest 3 decimal places]
- (e) $P(T < -2.660)$ for $n = 80$? [up to the nearest 3 decimal places]
- (f) $P(|T| > 1.363)$ for $df = 11$? [up to the nearest 1 decimal place]

3. Answer the following questions on-line by checking either **True** or **False**.

- (a) The t-distribution is characterized by the parameters μ (the mean) and σ (the standard deviation).
- (b) The t-table (Table D) works with the area above (right or upper tail) while the z-table (Table A) works with the area below (left or lower tail).
- (c) The t-distribution approaches the Standard Normal Distribution, $N(0, 1)$, as the degrees of freedom increase because s estimates σ with less precision when the sample size increases.

4. A study was conducted among individuals with chronic pain to determine how effective acupuncture is in relieving pain. A random sample of 15 subjects were chosen and their sensory rates were measured. Assume that the measurements on sensory rates follow a normal distribution. The data from the study are given below:

8.6 9.4 7.9 6.8 8.3 7.3 9.2 9.6 8.7 11.4 10.3 5.4 8.1 5.5 6.9

- (a) What is the value of $\bar{x}$, the sample mean (Use JMP or MS Excel to calculate)? (Round your answer to the nearest 2 decimal places)
- (b) What is the value of s , the sample standard deviation (Use JMP or MS Excel to calculate)? (Round your answer to the nearest 2 decimal places)
- (c) Report the degrees of freedom associated with the t-distribution.
- (d) What is the value of $t_{\frac{\alpha}{2}, n-1}$ for the 95% confidence interval for the mean sensory rate? (The value obtained from Table-D, accurate to the nearest 3 decimal places)
- (e) What is the margin of error for the 95% confidence interval for the mean sensory rate? (Round your answer to the nearest 2 decimal places)
- (f) Calculate the 95% confidence interval for the mean sensory rate.
 - i. What is the value of the lower bound of the 95% confidence interval? (Round your answer to the nearest 2 decimal places)
 - ii. What is the value of the upper bound of the 95% confidence interval? (Round your answer to the nearest 2 decimal places)
- (g) **[Free Response]** Interpret the confidence interval you constructed above in the context of the problem.
- (h) Assume that the standard deviation of all the sensory rates is known to be 1.67. Calculate the 95% confidence interval for the mean sensory rate under this situation.
 - i. What is the value of the lower bound of the 95% confidence interval? (Round your answer to the nearest 2 decimal places)
 - ii. What is the value of the upper bound of the 95% confidence interval? (Round your answer to the nearest 2 decimal places)
- (i) **[Free Response]** Is the 95% confidence interval constructed in part(f) **wider** or **narrower** than the 95% confidence interval in part(h)? Explain why.